

ALLIANCE HIGH SCHOOL

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Dear Admission Committee,

REF: RECOMMENDATION FOR SWAIB OKITA OUMA, ADMISSION NUMBER 11103

I had the immense pleasure of teaching Swaib Physics for his first three years and Mathematics for his final year. He set himself apart as exceptionally talented. Never failing to articulate difficult concepts, he came up with new ways of solving problems and this inspired a growth mindset in his peers. At the outset, Swaib was a shy, reserved boy. Gradually, he morphed out of this cocoon and by the second semester, his indomitable capacity for information seeking was unmistakable. I was astounded by his avid research on the disciplines we discussed in class. He made abstract connections between theories and their practicality, a feat I had never seen in any of my students. He actively challenged concepts, refusing to believe, like some traditional Physicists, theories that are presented with such certainty that they need not be substantiated experimentally. For instance, he refused to blindly accept *Ampere's Swimming Rule*. Due to his persistence, I was compelled to take the entire class to the Physics laboratory to demonstrate the rule.

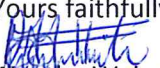
Seizing this opportunity, he made himself a recurring figure in the laboratory to the point the technician allowed him to carry out any type of research in the labs. As a result, his interest in Physics grew, and his imagination was boundless. It came as no surprise, therefore, when he came up with a spectacular science project whereby he tried to generate electricity from electrons as they flowed to neutralize charge on a positively charged electroscope. While the idea seemed achievable in theory, Swaib realized that in practice, he needed an instrument that could generate enough charge. He opted for the Van de Graaff generator, which the school did not have. Refusing to let this bar him, he wrote to Alliance Girls High School, our sister school, only to be told a day to the *Kenya Science and Engineering Fair* (KSEF) that their generator was also in use. This did not deter him either. He quickly picked himself up and gave a wonderful presentation on his project, despite knowing that he would not be evaluated since he lacked a prototype. A year later, when he went for an exchange program to *Brooks School, North Andover, MA*, he carried out the experiment. Though it failed, it was crucial in helping him understand, as he put it, that he needed a device that could hold more charge than an electroscope. In his final year, this experience was critical in helping his new project, titled 'Smart Water', win first place in the *KSEF*.

As a Mathematician, he was gritty and meticulous. His data analysis skills and interpretation of questions was unfathomable. His empathy and team-oriented thinking made him an invaluable asset to the weaker students. They always approached him when they had any trouble and he was more than willing to help them. In his second year, he wrote to the principal requesting for reforms in club positions that were previously solely held by seniors. In the successful proposal that collected a record 1300 signatures, he requested that all clubs include student leaders from all ages, to not only reduce the monopoly of senior students but also to give the younger students creative space to develop their ideas

and learn invaluable skills. This, coupled with his stellar performance, made him elected into the *Mathematics Club*. In this position, he was instrumental in reviving and organizing the *Carey Francis Mathematical Contest*, which was once a national classic. He played a pivotal role in ardently leading the team that participated in the *Kenya Mathematics Olympiad* where he was ranked 8th nationally.

Outside class, Swaib was a whole-new kettle of fish. He buzzed with enthusiasm while carrying out his duties with utmost diligence. His intrinsic entrepreneurial spirit, infused with his desire to help others, distinguishes him from the crowd. Right from his first year, he sold sweets to the school community, at 100% profit, and eventually employed other students as the business grew. This graduated into printing t-shirts for all school events and the school basketball and soccer teams. All the profit garnered went into raising money for social organizations such as the *Kenya Red Cross*. In 2015, he managed to raise an astonishing 1,670 USD that went into the annual *Mater Heart Run*, which funds children in need of cardiac surgeries. In his last year, he was appointed a *Christian Union Official*, doubling up as the Librarian and the Finance Officer. In this position, he changed the food items the club sold in order to bring in more profit, money that went towards welfare. He organized visits to the nearby *Kikuyu Hospital* and *PCEA Thogoto Old People's Home* where students gave back to the community by tending to the sick and elderly. Once in a while, they visited children's homes and donated items to them. Beyond high school, he has continued in this light by investing some of his savings into selling street grilled chicken, whose income is used in supporting small community projects.

I unreservedly recommend Swaib for admission to your acclaimed university. I strongly believe that his desire to innovate, expression of ideas, grounded leadership skills and amicable relations with his peers and teachers alike will be cemented by the all rounded education that your university provides. I am confident he will give back just as much.

Yours faithfully,

Nyuthe Wainaina

